

## Chapter 11. Sequences, Series, and Power Series

(수열, 급수, 거듭제곱급수)

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## § 11.1 Sequences(수열)

**Monotonic & Bounded : 개념 소개 및 적용**  
(Ex.12-13, Box 12 까지)

## ■ Infinite Sequences(무한수열)

An **infinite sequence**, or just a **sequence**, can be thought of as a list of numbers written in a definite order:

$$a_1, a_2, a_3, a_4, \dots, a_n, \dots$$

**NOTATION** The sequence  $\{a_1, a_2, a_3, \dots\}$  is also denoted by

$$\{a_n\} \quad \text{or} \quad \{a_n\}_{n=1}^{\infty}$$

Unless otherwise stated, we assume that  $n$  starts at 1.

**EXAMPLE 1** Some sequences can be defined by giving a formula for the  $n$ th term.

$$(a) \quad \left\{ \frac{1}{2^n} \right\} \quad a_n = \frac{1}{2^n} \quad \left\{ \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32}, \dots, \frac{1}{2^n}, \dots \right\}$$

$$(b) \quad \left\{ \frac{n}{n+1} \right\}_{n=2}^{\infty} \quad a_n = \frac{n}{n+1} \quad \left\{ \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{6}, \dots \right\}$$

$$(c) \quad \left\{ \sqrt{n+2} \right\}_{n=1}^{\infty} \quad a_n = \sqrt{n}, n \geq 3 \quad \{ \sqrt{3}, \sqrt{4}, \sqrt{5}, \sqrt{6}, \dots \}$$

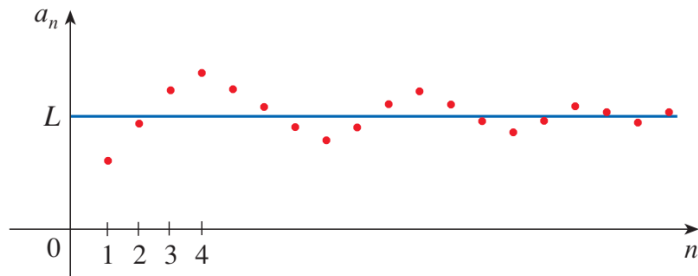
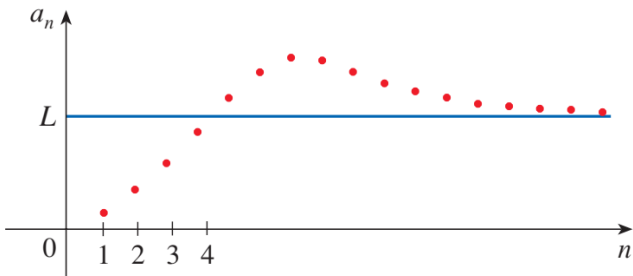
$$(d) \quad \left\{ (-1)^n \frac{(n+1)}{3^n} \right\}_{n=0}^{\infty} \quad \left\{ \frac{1}{1}, -\frac{2}{3}, \frac{3}{9}, -\frac{4}{27}, \frac{5}{81}, \dots \right\}$$

## ■ The Limit of a Sequence (수열의 극한)

**1 Intuitive Definition of a Limit of a Sequence** A sequence  $\{a_n\}$  has the **limit**  $L$  and we write

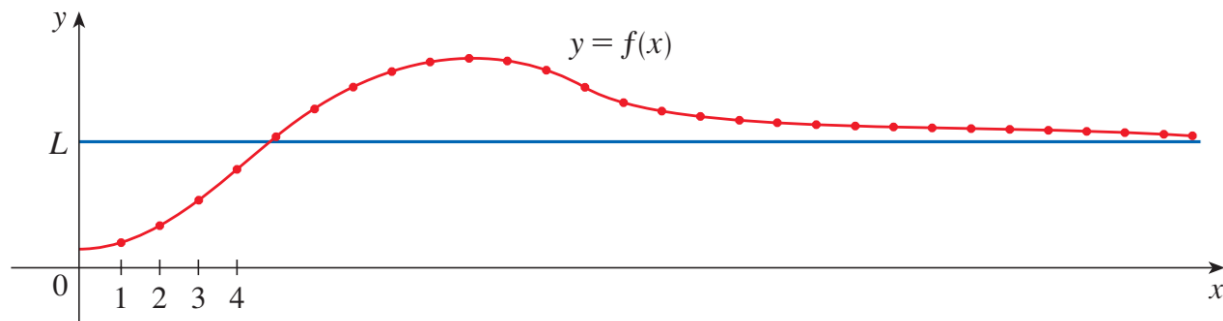
$$\lim_{n \rightarrow \infty} a_n = L \quad \text{or} \quad a_n \rightarrow L \text{ as } n \rightarrow \infty$$

if we can make the terms  $a_n$  as close to  $L$  as we like by taking  $n$  sufficiently large. If  $\lim_{n \rightarrow \infty} a_n$  exists, we say the sequence **converges** (or is **convergent**). Otherwise, we say the sequence **diverges** (or is **divergent**).



## ■ Properties of Convergent Sequences

**4 Theorem** If  $\lim_{x \rightarrow \infty} f(x) = L$  and  $f(n) = a_n$  when  $n$  is an integer, then  $\lim_{n \rightarrow \infty} a_n = L$ .



**Limit Laws for Sequences** Suppose that  $\{a_n\}$  and  $\{b_n\}$  are convergent sequences and  $c$  is a constant. Then

$$1. \lim_{n \rightarrow \infty} (a_n + b_n) = \lim_{n \rightarrow \infty} a_n + \lim_{n \rightarrow \infty} b_n$$

$$2. \lim_{n \rightarrow \infty} (a_n - b_n) = \lim_{n \rightarrow \infty} a_n - \lim_{n \rightarrow \infty} b_n$$

$$3. \lim_{n \rightarrow \infty} ca_n = c \lim_{n \rightarrow \infty} a_n$$

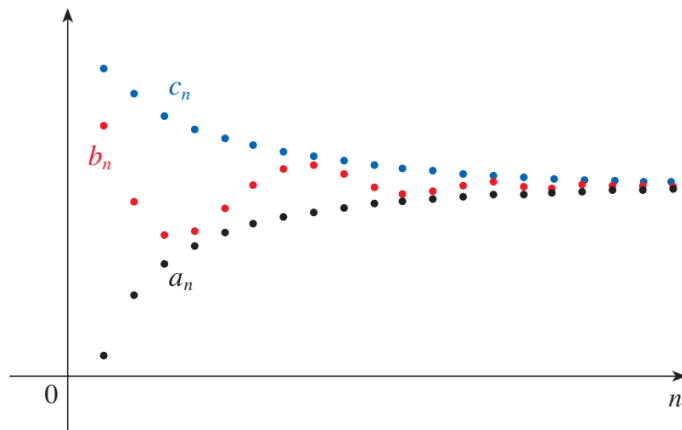
$$4. \lim_{n \rightarrow \infty} (a_n b_n) = \lim_{n \rightarrow \infty} a_n \cdot \lim_{n \rightarrow \infty} b_n$$

$$5. \lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{\lim_{n \rightarrow \infty} a_n}{\lim_{n \rightarrow \infty} b_n} \quad \text{if } \lim_{n \rightarrow \infty} b_n \neq 0$$

$$\lim_{n \rightarrow \infty} a_n^p = \left[ \lim_{n \rightarrow \infty} a_n \right]^p \quad \text{if } p > 0 \text{ and } a_n > 0$$

The Squeeze Theorem ( **for Sequences** ), 압축정리

If  $a_n \leq b_n \leq c_n$  for  $n \geq n_0$  and  $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} c_n = L$ , then  $\lim_{n \rightarrow \infty} b_n = L$ .



**6 Theorem**

If  $\lim_{n \rightarrow \infty} |a_n| = 0$ , then  $\lim_{n \rightarrow \infty} a_n = 0$ .

**EXAMPLE 6** Calculate  $\lim_{n \rightarrow \infty} \frac{\ln n}{n}$ .

**Solution:**

**7 Theorem** If  $\lim_{n \rightarrow \infty} a_n = L$  and the function  $f$  is continuous at  $L$ , then

$$\lim_{n \rightarrow \infty} f(a_n) = f(L)$$

**EXAMPLE 9** Find  $\lim_{n \rightarrow \infty} \sin \frac{\pi}{n}$ .

**Solution:**

**9** The sequence  $\{r^n\}$  is convergent if  $-1 < r \leq 1$  and divergent for all other values of  $r$ .

$$\lim_{n \rightarrow \infty} r^n = \begin{cases} 0 & \text{if } -1 < r < 1 \\ 1 & \text{if } r = 1 \end{cases}$$

## ■ Monotonic and Bounded Sequences (단조유계수열)

**10 Definition** A sequence  $\{a_n\}$  is called **increasing** if  $a_n < a_{n+1}$  for all  $n \geq 1$ , that is,  $a_1 < a_2 < a_3 < \dots$ . It is called **decreasing** if  $a_n > a_{n+1}$  for all  $n \geq 1$ . A sequence is called **monotonic** if it is either increasing or decreasing.

**EXAMPLE 12** The sequence  $\left\{ \frac{3}{n+5} \right\}$

**EXAMPLE 13** Show that the sequence  $a_n = \frac{n}{n^2 + 1}$  is decreasing.

**Solution:**

**11 Definition** A sequence  $\{a_n\}$  is **bounded above** if there is a number  $M$  such that  
(위로 유계)

$$a_n \leq M \quad \text{for all } n \geq 1$$

A sequence is **bounded below** if there is a number  $m$  such that  
(아래로 유계)

$$m \leq a_n \quad \text{for all } n \geq 1$$

If a sequence is bounded above and below, then it is called a **bounded sequence**.  
(유계 수열)

(단조수열정리)

**12 Monotonic Sequence Theorem** Every bounded, monotonic sequence is convergent.

In particular, a sequence that is increasing and bounded above converges, and a sequence that is decreasing and bounded below converges.

